

GuideMate

Version 1.0

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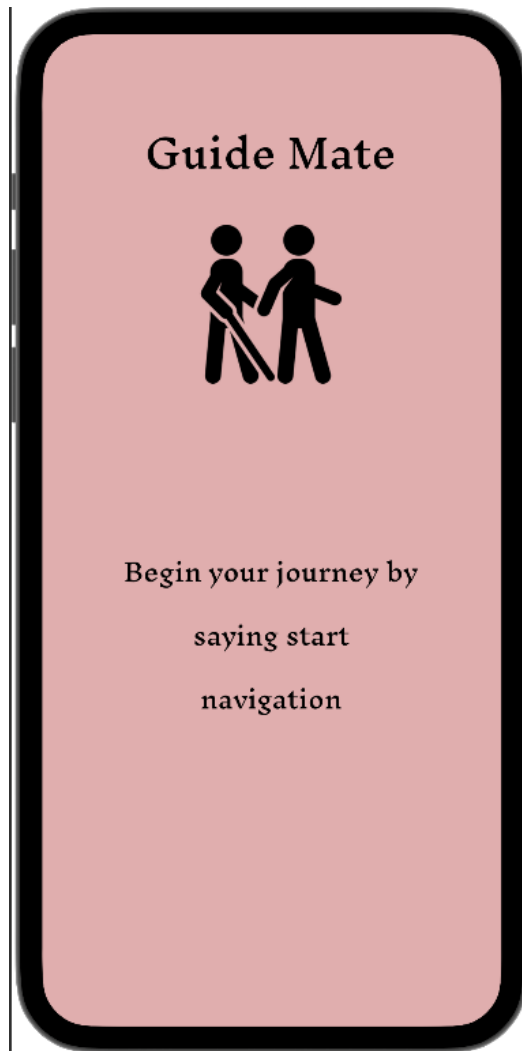
Introduction

Thanks for choosing to use GuideMate! We aim to satisfy all your safe, indoor navigation needs within ECSW, ECSN, or ECSS inside the UTD campus

This user manual serves as a comprehensive guide to using GuideMate, as well as an overview of the app's available features.

"To boldly navigate safely where no one has gone before"™

The Landing Page



Upon loading the app, you will be presented with the home screen as shown above.

GuideMate is designed for accessibility to accommodate for users with visual impairments. The app primarily supports voice navigation; thus, to begin navigation, the user must say the specific phrase shown above: "Start Navigation"

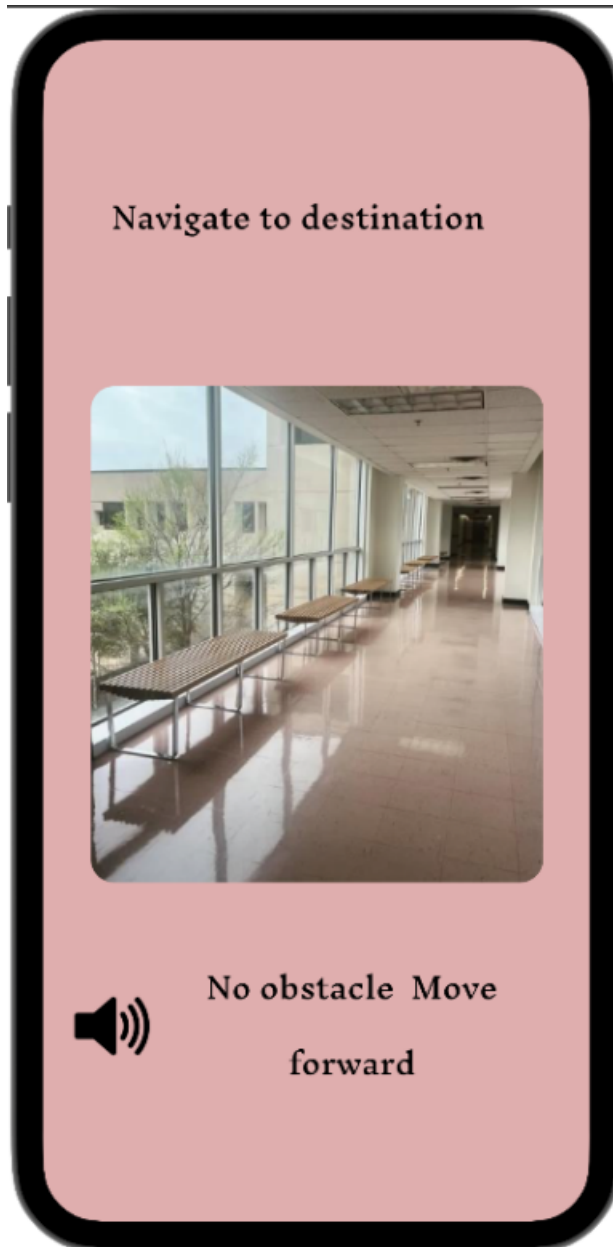
The Camera



GuideMate does not just statically navigate the user along a route towards the destination. Rather, to ensure safe navigation, the app also offers obstacle detection and warning systems, which is accomplished using the device's built-in camera.

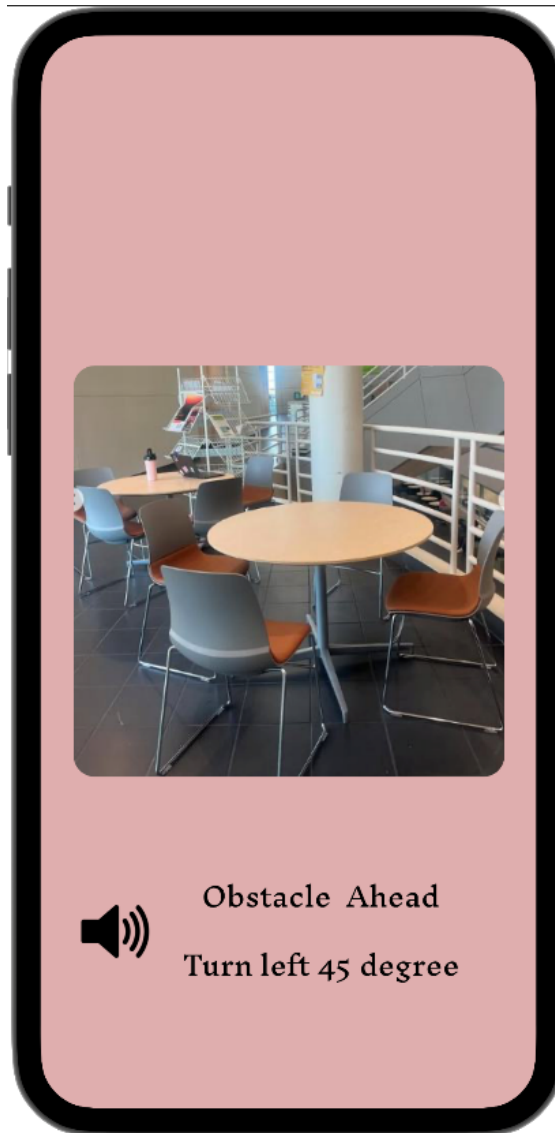
The app prompts the user to enable camera access to the app if it is not already allowed. Otherwise, the app will prompt the user to point their camera in front of them, acting as a second pair of eyes. A live video view on the app represents what the camera currently sees.

Navigation: No Obstacles



When no obstacles are detected, the app simply instructs the user to move normally.

Navigation: Obstacles Detected



With a destination selected, GuideMate begins to detect objects and issue navigation instructions. As pictured above, the app detects students sitting down as obstacles, and instructs the user to turn 45 degrees to the Left. GuideMate uses explicit measurements to reduce ambiguity.

Navigation: Completed



Once GuideMate identifies that the user has reached the destination location, it notifies the user via text-to-speech and terminates navigation.

Emergency

No system is foolproof, and accidents or emergency situations may occur.

When GuideMate detects a fall or loses the ability to pinpoint the current location, the app will prompt the user to call for emergency services. This gives the user an opportunity to deny the request in case emergency services is not needed, such as if they had merely dropped the phone for a moment.

However, if the user agrees or remains unresponsive after a short amount of time, GuideMate will proceed to place calls and messages to emergency services and the saved emergency contact. These calls and messages contain the app's best approximation of the user's current location, and includes other information such as the user possibly being unresponsive.

Prior to beginning navigation, the user is prompted to save an emergency contact. If one is not provided, the app will only call and message emergency services like 911.

Customization

GuideMate accommodates the user's individual preferences by offering various customization options.

These options include changing the volume level of the app's vocalizations and the time interval between these vocalizations.

The user can apply conditions to their navigation routes, such as indicating a preference to avoid staircases. The app will attempt to find a safe and efficient route under these conditions, or indicate if no such route is available.